

Mapping Invasive *Spartina alterniflora* Using Phenological Information and Red-Edge Bands of Sentinel-2 Time-Series Data

Yiwei Ma , Li Zhuo , and Jingjing Cao 

Abstract—The accurate mapping of *Spartina alterniflora* (*S. alterniflora*) invasion is crucial for controlling its spread and reducing severe ecological problems. Satellite images have been extensively employed for *S. alterniflora* invasion monitoring; however, there are still several issues that need to be addressed. The spectral similarities between *S. alterniflora* and surrounding ground objects make it challenging for traditional classifiers to achieve satisfactory extraction accuracy. Since the phenological information and red-edge spectral differences have been considered as informative features for identifying *S. alterniflora*, current studies mainly used them separately as classification features and seldom considered the differences of red-edge information at different phenological periods. Therefore, we proposed a pixel-based phenological and red-edge feature composite method (PpRef-CM) for *S. alterniflora* extraction considering both phenological information and red-edge bands derived from Sentinel-2 time series based on the existing pixel-based phenological feature composite method (Ppf-CM). The proposed PpRef-CM and machine-learning algorithms were employed for *S. alterniflora* extraction in two typical mangrove forests along coastal China. Results indicated that red-edge information at different phenological periods is essential for detecting *S. alterniflora*. *S. alterniflora* extraction achieved the highest accuracy of 96.57% by using the eXtreme gradient boost algorithm when compared with other machine-learning algorithms. The PpRef-CM gave 2.72% and 2.61% more extraction accuracies of *S. alterniflora* than the Ppf-CM in two study sites, separately. These findings provide insights for selecting suitable classification features for *S. alterniflora* extraction studies and serve as an effective control and management of *S. alterniflora*.

Index Terms—Invasive species, phenology, red-edge bands, Sentinel-2 time series, *Spartina alterniflora* (*S. alterniflora*).

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I. INTRODUCTION

SPARTINA *alterniflora* (*S. alterniflora*) was generally considered as an effective means of sediment decomposition and coastal protection in the early stages since it was introduced to China in 1979 [1], [2]. Subsequently, *S. alterniflora* expanded expeditiously along the coast of China owing to its adaptability and tolerance for the hypoxic and high-salinity environment [3]. By 2020, the area of *S. alterniflora* in China reached 51970 ha, including 984 and 3398 ha of which had been separately converted from mangrove forests and mudflats since 2015 [4]. To date, *S. alterniflora* has been considered as an invasive species of great concern in wetland ecosystems along coastal China [3], [5], [6]. The rapid growth and expansion of *S. alterniflora* have several negative impacts on coastal wetlands, such as the replacement of native species, the habitat loss and food reduction of migratory birds, and the decrease of biodiversity [7], [8], [9].

The coastal wetlands are favorable habitats for numerous coastal vegetation, especially *S. alterniflora* and mangrove forests, possessing an extensive geographic range and complex geographic environments [10]. However, conventional field investigations are limited to large-scale monitoring of *S. alterniflora* due to their inaccessibility and inefficiency, etc. [11]. Remote sensing techniques with the advantages of wide-range, time-series and dynamic monitoring have been effectively used for *S. alterniflora* invasion monitoring [12], [13], [14], [15]. Over the past 10 years, Sentinel-2 has received more attention for its good performance in vegetation monitoring, when compared with other optical satellites [16]. It offers additional red-edge bands that can effectively capture the unique spectral characteristics of vegetation canopy [17], [18], and they have been explored in detecting *S. alterniflora* [19], [20].

Subject to the spectral similarities of *S. alterniflora* and surrounding vegetation and the intrinsic spectral variability within *S. alterniflora*, the issues of misclassification between *S. alterniflora* and surrounding ground objects are still not well handled, especially mangrove forests and mudflats [21], [22], [23]. Subsequently, the phenological information has been investigated in *S. alterniflora* invasion monitoring [24], [25], including the commonly used image-based multitemporal methods [26], [27] and the pixel-based image compositing methods [22], [28], [29], [30], [31]. Among them, the representative pixel-based phenological feature composite method (Ppf-CM) has proven to be effective in distinguishing *S. alterniflora* and

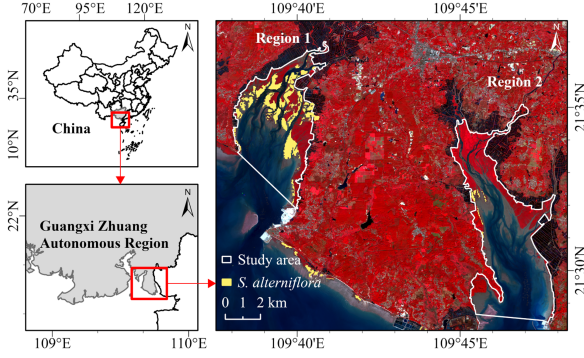


Fig. 1. Location of the Dandou Sea (Region 1) and the Yingluo Bay (Region 2) along coastal China, and the distribution of *S. alterniflora* at 30 m spatial resolution in 2015 [41], [42].

surrounding vegetative classes, which investigated the spectral indices, enhanced vegetation index, and plant senescence reflectance index (PSRI), at two phenological stages to generate the Landsat composite images in the green period and senescence period, and then employed the Ppf-CM based images to extract *S. alterniflora* [31]. Previous studies indicated that the spectrum of vegetation is sensitive to chlorophyll contents, especially in the red-edge region [32], [33], which have more significant variations at different phenological trajectories [34]. However, only conventional spectral band-derived indices have been considered in most pixel-based image compositing methods, the red-edge spectral indices during different phenological periods in extracting *S. alterniflora* are rarely explored.

Therefore, this study developed a new pixel-based phenological and red-edge feature composite method (PpRef-CM) for *S. alterniflora* extraction by introducing red-edge spectral indices of Sentinel-2 images based on the existing Ppf-CM method. Compared with the Ppf-CM, the proposed PpRef-CM generated the composite image in the growing season (GS) of *S. alterniflora* by using the red-edge spectral index rather than the conventional spectral index. The main objectives were to explore the effectiveness of the red-edge information of Sentinel-2 for *S. alterniflora* extraction and enhance the accuracy of *S. alterniflora* extraction by combining phenological information and red-edge bands of Sentinel-2 time series.

II. STUDY AREA AND DATASETS

A. Study Area

This study selected two representative mangrove forests (Dandou Sea and Yingluo Bay) located in the Shankou Mangrove National Nature Reserve, Guangxi, China (21°24'N to 21°37'N, 109°32'E to 109°51'E) [22], as our study area (see Fig. 1). The Dandou Sea belongs to a humid subtropical climate, with the average annual temperature and precipitation of 22.5 ± 0.5 °C and 2000 mm, separately, that circa 80% of precipitation concentrates during April to September [35]. The Yingluo Bay has a typical subtropical oceanic monsoon climate with an average annual precipitation of 1700 mm [36]. Based on a series of field investigations and unmanned aerial vehicle monitoring, there are four major classes of ground objects in the study area, including mangrove forests, *S. alterniflora*, mudflats, and

TABLE I
TRAINING SAMPLES AND TESTING SAMPLES OF *S. ALTERNIFLORA* AND OTHER THREE CLASSES IN THE STUDY AREA

Class name	Training samples	Testing samples	Total
<i>S. alterniflora</i>	493	210	703
Mangrove forests	487	219	706
Mudflat	503	204	707
Water	491	213	704
Total	1974	846	2820

water bodies. The dominant mangrove species in the study area include *Rhizophora stylosa*, *Bruguiera gymnorhiza*, *Aegiceras corniculatum*, and *Avicennia marina* [37]. *S. alterniflora* is a salt marsh grass, which has grown and spread rapidly in the study area since it was first introduced in 1979, and the area has reached 454.24 hm² in 2019 [38], [39]. Here, *S. alterniflora* was generally found in the intertidal zone of coastal mudflats, including the bare mudflats, the low tidal margins of mangrove forests, and the middle of canopy opening mangrove forests [39].

B. Data and Preprocessing

1) *Sentinel-2 Data*: This study selected the Sentinel-2 multispectral instrument (MSI) images based on the Google Earth Engine cloud platform for *S. alterniflora* mapping. The Sentinel-2 MSI Level-2A images were orthorectified and atmospherically corrected. Sentinel-2 has the advantages of a short revisit time of every 5 days and 13 spectral bands [28]. Considering the influences of frequent clouds and periodic tidal fluctuation [40], there are 26 Sentinel-2 images covering the Dandou Sea and 40 Sentinel-2 images covering the Yingluo Bay from 2019 to 2022 that were selected in this study, respectively. The cloud masking and abnormal value removal were then performed for these Sentinel-2 images. Besides, the QA60 band was utilized to block out the pixels of opaque and cirrus clouds in the Sentinel-2 images [22]. There were 11 spectral bands of Sentinel-2 that were employed in this study, including blue, green, and red bands, red-edge bands, near-infrared (NIR) band, water vapor (WV) band, and shortwave infrared (SWIR) bands. The spatial resolutions of these spectral bands were all resampled to 10 m by the bilinear interpolation method.

2) *Ground Reference Data*: Based on several field surveys conducted in November 2021, April and November 2023, and July 2024 in the study area, this study selected 2820 samples as the ground reference datasets of *S. alterniflora* and other ground objects from the Google Earth images using visual interpretation. Besides, a distribution map of *S. alterniflora* in China in 2015 at 30-m resolution provided by previous studies was served as a reference for sample selection [41], [42]. Eventually, there are 70% samples randomly selected for training machine-learning models and the rest selected as testing samples for accuracy evaluation (see Table I).

III. METHODOLOGY

This study extracted the distributions of *S. alterniflora* by integrating phenological information and red-edge band indices includes four steps (see Fig. 2). First, the Sentinel-2-derived

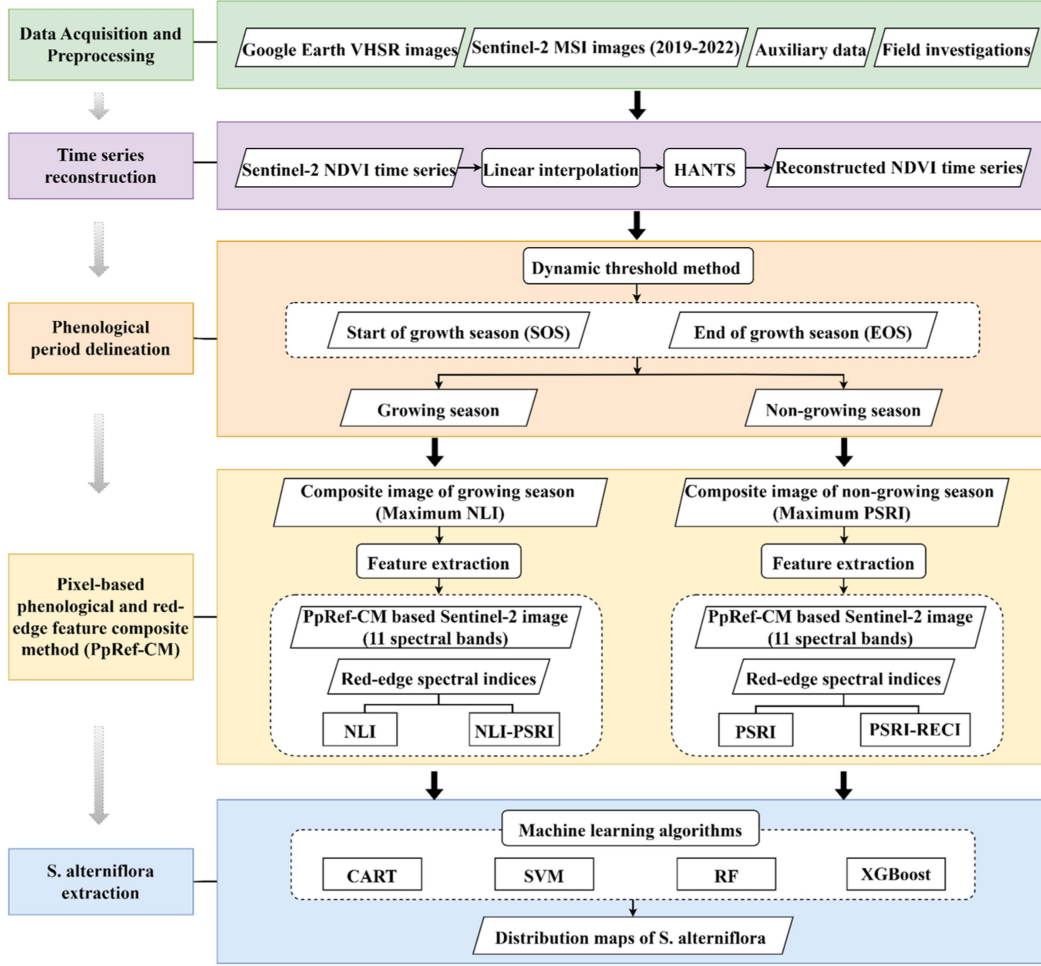


Fig. 2. Flowchart of the extraction for *S. alterniflora* by integrating phenological information and red-edge band indices of Sentinel-2 time series.

normalized difference vegetation index (NDVI) time series was reconstructed by using linear interpolation and harmonic analysis of NDVI time-series (HANTS) method [43], and two key phenological parameters, i.e., start of growth season (SOS) and end of growth season (EOS), were calculated by dynamic threshold method. Second, the composite images of nonlinear index (NLI) in the GS and the composite images of PSRI in the nongrowing season (NGS) were generated by using the maximum value composite method. Third, the spectral band features and red-edge band indices were obtained from these composite images. Finally, these features and machine-learning algorithms were employed to map the distribution of *S. alterniflora*.

A. Reconstruction of Sentinel-2 NDVI Time Series

To diminish the influence of noises and missing values in the Sentinel-2 images, linear interpolation and HANTS were adopted for data gap filling and fitting, respectively. The HANTS is a time-honored reconstruction method for the processing of satellite time-series images, and it has been popularly used in vegetation phenology studies [44]. Taking the commonly used NDVI as an example, as shown in (1), the HANTS treats a fitted NDVI curve as the sum of a set of cosine waves and an additive

term, by which the original NDVI time series can be decomposed into an infinite series of harmonic components [43]

$$y(t_j) = a_0 + \sum_{i=1}^{nf} [a_i \cos(2\pi f_i t_j) + b_i \sin(2\pi f_i t_j)] \quad (1)$$

where a_0 denotes the coefficient at zero frequency, b_i denotes the coefficient of a trigonometric component with frequencies f_i , and t_j denotes the imaging time of time-series data.

B. Delineating the GS and NGS of *S. Alterniflora*

The spectrum of *S. alterniflora* differs significantly in the GS and NGS [14], which helps increase the separability of *S. alterniflora* between other ground objects. To effectively utilize the phenological information, the GS and NGS of *S. alterniflora* were delineated by two key phenological parameters (SOS and EOS). Specifically, the dates between SOS and EOS were defined as the GS, while the rest of the year as the NGS, referring to the existing studies [26], [31], [45]. The SOS and EOS were estimated by using the Sentinel-2 NDVI time series and dynamic threshold method [46], [47]. The calculations of the NDVI-derived SOS and EOS (NDVI_{SOS} and NDVI_{EOS}) are

TABLE II
SPECTRAL BAND FEATURES OF SENTINEL-2 IN TWO PHENOLOGICAL PERIODS

Phenological period	Spectral band features
GS	GS_Blue, GS_Green, GS_Red, GS_RE1, GS_RE2, GS_RE3, GS_NIR, GS_RE4, GS_WV, GS_SWIR1, and GS_SWIR2
NGS	NGS_Blue, NGS_Green, NGS_Red, NGS_RE1, NGS_RE2, NGS_RE3, NGS_NIR, NGS_RE4, NGS_WV, NGS_SWIR1, and NGS_SWIR2

Note: Blue, Green, Red, RE1, RE2, RE3, RE4, NIR, WV, SWIR1, and SWIR2 denote the blue, green, and red bands (Bands 2–4), red-edge bands (Bands 5–7 and 8A), NIR band (Band 8), water vapor band (Band 9), and SWIR bands (Bands 11 and 12) of Sentinel-2, respectively; GS_XXX and NGS_XXX denote a particular spectral band (XXX) in the GS and the NGS, respectively.

TABLE III
DEFINITIONS AND CALCULATIONS OF FOUR RED-EDGE BAND INDICES USED IN THIS STUDY

Red-edge spectral index	Calculation	Reference
NLI	$\frac{\rho_{RE3}^2 - \rho_{Red}}{\rho_{RE3}^2 + \rho_{Red}}$	[49]
PSRI	$\frac{\rho_{RE1}}{\rho_{Red} - \rho_{Blue}}$	[50]
The difference between NLI and PSRI (NLI-PSRI)	$\frac{\rho_{RE3}^2 - \rho_{Red}}{\rho_{RE3}^2 + \rho_{Red}} - \frac{\rho_{Red} - \rho_{Blue}}{\rho_{RE1}}$	This study
The difference between PSRI and RECI (PSRI-RECI)	$\frac{\rho_{RE1}}{\rho_{Red} - \rho_{Blue}} - \left(\frac{\rho_{NIR}}{\rho_{RE1}} - 1 \right)$	This study

Note: ρ_{Blue} , ρ_{Red} , ρ_{RE1} , ρ_{RE3} , and ρ_{NIR} represent the reflectance in the blue band (Band 2), red band (Band 4), red-edge bands (Bands 5 and 7), and NIR band (Band 8) of Sentinel-2, respectively.

as follows:

$$NDVI_{\text{SOS}} = NDVI_{\text{leftmin}} + 0.2 \times (NDVI_{\text{max}} - NDVI_{\text{leftmin}}) \quad (2)$$

$$NDVI_{\text{EOS}} = NDVI_{\text{rightmin}} + 0.5 \times (NDVI_{\text{max}} - NDVI_{\text{rightmin}}) \quad (3)$$

where $NDVI_{\text{leftmin}}$ and $NDVI_{\text{rightmin}}$ denote the minimums on the left and right sides of NDVI time-series curve, separately; $NDVI_{\text{max}}$ denotes the maximum of NDVI time-series curve.

C. Remote Sensing Feature Extraction Considering Different Phenological Periods

Considering the difference in spectral properties of different phenological periods and the effectiveness of red-edge bands in distinguishing wetland vegetation classes [48], this study developed a PpRef-CM to fill the gap in existing Ppf-CM for accurate *S. alterniflora* extraction. Specifically, the maximum value composite image of NLI in the GS and the maximum value composite image of PSRI in the NGS were constructed by using Sentinel-2 time series. There are a total of 26 spectral features selected from these composite images in two phenological periods of a year for *S. alterniflora* extraction (see Tables II and III), including 22 spectral band features and four red-edge spectral indices. The NLI can fully utilize the unique characteristics of vegetation in the red-edge region of spectrum [49], and the PSRI is sensitive to leaf senescence-induced reflectance changes [50]. Referring to an existing study [51], to highlight the difference between *S. alterniflora* and other ground objects, we integrated the spectral indices by conducting a subtraction operation between NLI and PSRI, and between PSRI and red-edge chlorophyll index (RECI) [52].

D. Machine-Learning-Based Extraction and Precision Evaluation

The classification performance could be related to the learning pattern of a specific algorithm and the contribution of each classification feature [53]. Advanced machine-learning algorithms have been increasingly applied for vegetation classification, due to their capabilities in handling complex and high-dimensional data [54], [55], [56], [57], [58]. Recently, eXtreme gradient boost (XGBoost) algorithm has demonstrated advantages in remote sensing classification [59], [60]. It is a scalable machine-learning algorithm in the gradient boost framework and is suitable for handling the issue of sparse data [61]. The addition of a regularization parameter in the objective function of XGBoost prevents overfitting by constraining the complexity of the regression tree functions. Since the ensemble process of XGBoost holds the capabilities of error smoothing [62], it is unsusceptible to the completed correlated data in multifeature classification [63].

To verify the effectiveness of XGBoost on *S. alterniflora* extraction, three popular machine-learning algorithms were also employed, i.e., classification and regression tree (CART) [64], support vector machine (SVM) [65], and random forest (RF) [66]. Among them, CART is an effective prediction algorithm that partitions the instances into subsets with homogenous values of the dependent variable; however, it is susceptible to minor changes in the data [67]. SVM is a kernel-based algorithm that constructs the optimal hyperspace in the feature space to separate samples, and it has prominent adaptability to nonlinear data [68]. RF is insensitive to the outliers by ensembling multiple decision trees under the bagging framework [69], [70]. The CART, SVM, RF, and XGBoost were utilized for the extraction of *S. alterniflora*. The parameters of these machine-learning algorithms were optimized using the GridSearchCV function [71].

TABLE IV
PHENOLOGICAL PARAMETERS AND PERIODS OF *S. ALTERNIFLORA* IN OUR STUDY AREA

Year	Phenological parameter		Phenological period	
	SOS	EOS	GS	NGS
2019	DOY 112	DOY 322	DOY 112–322	DOY 1–111, DOY 323–365
2020	DOY 117	DOY 322	DOY 117–322	DOY 1–116, DOY 323–365
2021	DOY 116	DOY 321	DOY 116–321	DOY 1–115, DOY 322–365
2022	DOY 116	DOY 321	DOY 116–321	DOY 1–115, DOY 322–365

TABLE V
EXTRACTION ACCURACY OF *S. ALTERNIFLORA* AND OTHER GROUND OBJECTS USING MACHINE-LEARNING ALGORITHMS

Class	CART		SVM		RF		XGBoost	
	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)
<i>S. alterniflora</i>	93.33	92.02	96.19	96.19	94.76	95.67	96.19	94.39
Mangrove forests	94.06	97.63	95.43	98.58	97.26	99.53	97.26	99.53
Mudflats	91.67	89.05	96.08	91.59	96.08	92.02	94.61	93.69
Water	96.71	97.17	97.65	99.05	97.65	98.58	98.12	98.58
OA (%)	93.97		96.34		96.45		96.57	
Kappa coefficient	0.9196		0.9511		0.9527		0.9543	

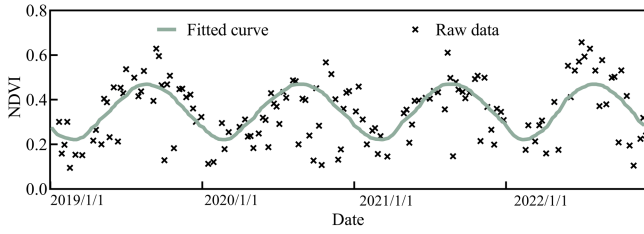


Fig. 3. Example of the fitted curve and raw data derived from Sentinel-2 NDVI time series.

To quantify the performance of machine-learning algorithms employed for extracting *S. alterniflora*, four indicators, including the overall accuracy (OA), Kappa coefficient, producer's accuracy (PA), and user's accuracy (UA), were selected and calculated by using confusion matrices and samples [28], [72].

IV. RESULTS

A. Phenological Parameter Extraction of *S. Alterniflora* Based on Sentinel-2 Time Series

To investigate the phenological characteristics of *S. alterniflora*, Fig. 3 shows an example of the fitted curve derived from time series of Sentinel-2 NDVI. It can be seen that the NDVI started to rise in February, peaked around late August, and turned to decline in early September, indicating that *S. alterniflora* has obvious phenological characteristics and it is consistent with an existing study [27]. The SOS and EOS of *S. alterniflora* in the study area during 2019–2022 were obtained by using NDVI time series and dynamic threshold method (see Table IV), which corroborates the findings of previous studies [14], [27]. Subsequently, using the SOS and EOS as a basis, the phenological period of *S. alterniflora* in the study area was split into the GS and NGS.

To visually analyze the spectral differences in two phenological periods, the reflectance curves of *S. alterniflora* and other

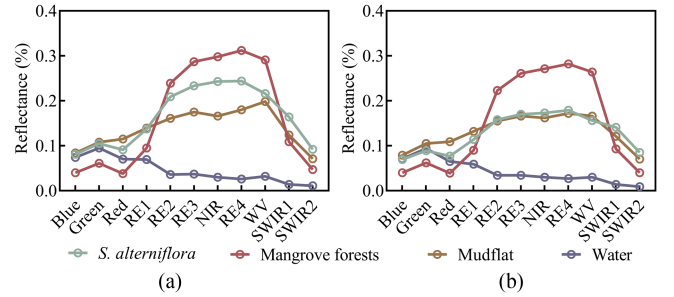


Fig. 4. Reflective spectral curves of four classes in the study area in the (a) GS and (b) NGS.

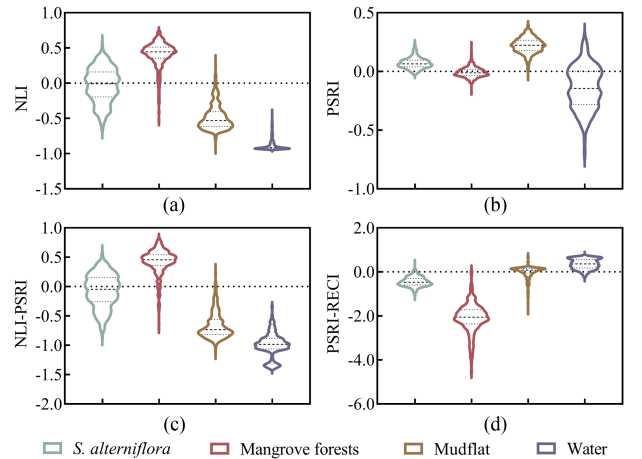


Fig. 5. Violin plots of different red-edge band indices. (a) NLI, (b) PSRI, (c) NLI-PSRI, and (d) PSRI-RECI for *S. alterniflora* and other three classes.

three classes were shown in Fig. 4. Fig. 4(a) indicates that the obvious reflectance differences existed between *S. alterniflora* and mudflats in the GS among the RE2, RE3, and RE4 bands of Sentinel-2. Fig. 4(b) also demonstrates that the reflectance curve

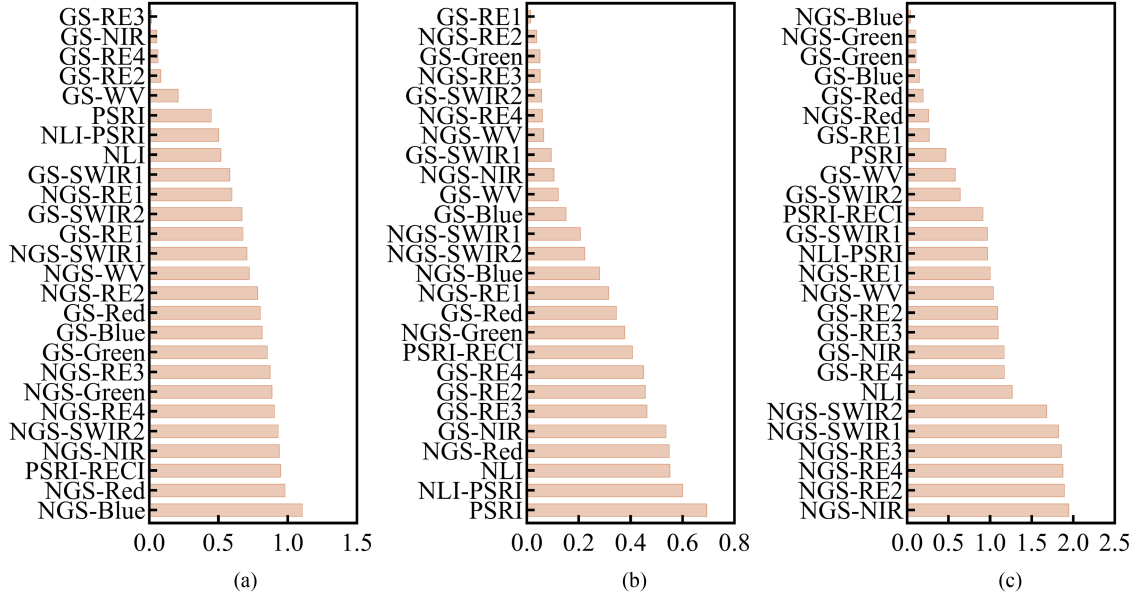


Fig. 6. SI values of classification features for separating (a) *S. alterniflora* and mangrove forests, (b) *S. alterniflora* and mudflats, and (c) *S. alterniflora* and water.

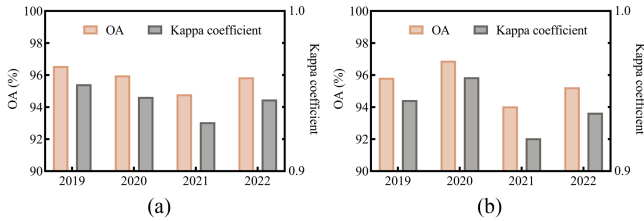


Fig. 7. Extraction accuracy of *S. alterniflora* in the Dandou Sea (Region 1) and the Yingluo Bay (Region 2) from 2019 to 2022.

of *S. alterniflora* was distinct from that of mangrove forests in the NGS. These findings indicated that the phenological differences in the red-edge bands can help to distinguish *S. alterniflora* from other classes, such as mangrove forests and mudflats.

B. Extraction of *S. Alterniflora* Using Phenological and Red-Edge Information

The extraction accuracies of *S. alterniflora* were obtained by using the CART, SVM, RF, and XGBoost along with all the classification features, as shown in Table V. As for these machine-learning algorithms, the *S. alterniflora* extraction accuracies of them were all higher than 93%. The XGBoost outperformed the other three algorithms and got the highest accuracy of 96.57% and the Kappa coefficient of 0.9543.

To compare the ability of different red-edge spectral indices for *S. alterniflora* extraction, the data distributions of these indices for four classes were shown in Fig. 5. As can be seen, *S. alterniflora* could be accurately separated from other classes based on these red-edge spectral indices. The range distributions of NLI, PSRI, and NLI-PSRI between *S. alterniflora* and mudflats [see Fig. 5(a)–(c)] and the range distribution of PSRI-RECI between *S. alterniflora* and mangrove forests [see Fig. 5(d)] were

TABLE VI
EXTRACTION ACCURACY OF *S. ALTERNIFLORA* WITH OR WITHOUT RED-EDGE INFORMATION

Feature setting	OA (%)	Kappa coefficient
A	95.74	0.9432
A+B	95.98	0.9464
A+C	96.45	0.9527
A+B+C (this study)	96.57	0.9543

Note: A denotes the feature set, including blue, green, red, NIR, water vapor, and SWIR bands of PpRef-CM-based Sentinel-2 images; B and C denote the feature sets, including red-edge bands and red-edge spectral indices of PpRef-CM-based Sentinel-2 images, respectively.

distinctly different; therefore, these red-edge spectral indices' features could be useful for *S. alterniflora* extraction.

To explore the validity of red-edge information for *S. alterniflora* extraction, four feature sets were considered separately for ablation studies. As shown in Table VI, compared with the conventional spectral bands (feature set A), the inclusion of red-edge bands (feature set B) or red-edge spectral indices (feature set C) derived from PpRef-CM-based Sentinel-2 images can further enhance the extraction accuracies. However, all the classification features (feature sets A–C) gave the best performance of *S. alterniflora* extraction.

To further verify the effectiveness of the red-edge information, the separability index (SI) was employed to evaluate the performance of different classification features. The SI denotes the separability between two classes by considering their intra-class and interclass variabilities, and the larger SI indicates the greater separability of different classes [9], [73]. The SI can be calculated as follows:

$$SI_{sj} = \frac{\Delta_{inter_{sj}}}{\Delta_{intra_{sj}}} = \frac{|\overline{\mu_s} - \overline{\mu_j}|}{1.96 \times (\sigma_s - \sigma_j)} \quad (4)$$

where $\overline{\mu_s}$ and σ_s represent the mean and standard deviation (SD) of classification features of *S. alterniflora*, respectively; $\overline{\mu_j}$ and

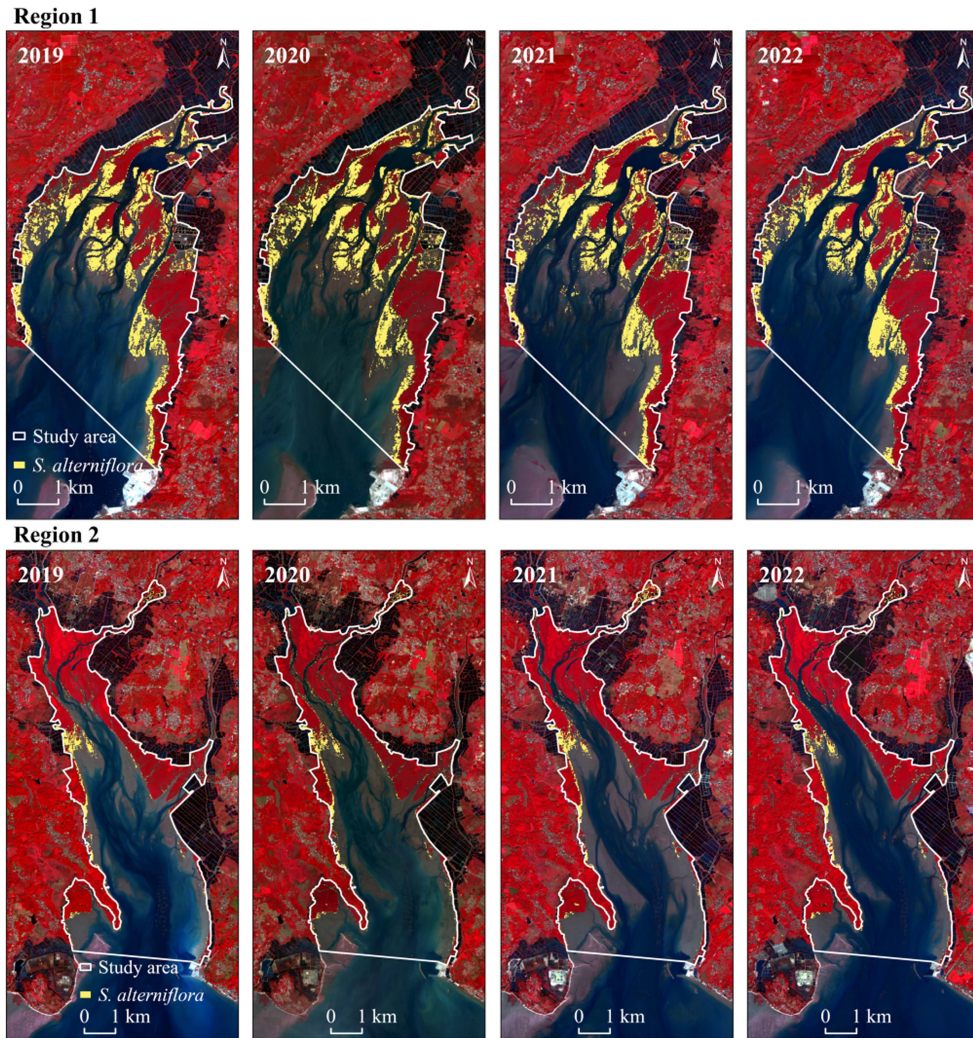


Fig. 8. Distribution maps of *S. alterniflora* in the Dandou Sea (Region 1) and the Yingluo Bay (Region 2) from 2019 to 2022.

σ_j represent the mean and SD of classification features of other three classes, respectively.

The ranking of SI for classification features demonstrated that the separability of red-edge bands varied in two phenological periods (see Fig. 6). The red-edge bands of the GS (i.e., GS-RE3, GS-RE2, and GS-RE4) are crucial for separating *S. alterniflora* from mudflats, while the red-edge bands of the NGS (i.e., NGS-RE4, NGS-RE3, and NGS-RE2) contribute to discriminating *S. alterniflora* from mangrove forests. Moreover, the red-edge spectral indices derived from PpRef-CM-based Sentinel-2 images were helpful in the extraction of *S. alterniflora*. The high SI value of PSRI-RECI indicated its excellent ability to separate *S. alterniflora* and mangrove forests [see Fig. 6(a)]. The PSRI, NLI-PSRI, and NLI had high SI values in separating *S. alterniflora* and mudflats [see Fig. 6(b)].

C. Distribution Maps of *S. Alterniflora* in the Study Area

The performances of red-edge bands and red-edge spectral indices derived from PpRef-CM-based Sentinel-2 images for

extracting *S. alterniflora* were evaluated. Fig. 7 shows the extraction accuracies of *S. alterniflora* in two study sites (Regions 1 and 2) from 2019 to 2022 obtained by the XGBoost, with the overall accuracies of each year higher than 94% and the Kappa coefficients higher than 0.92.

Based on the distribution maps of *S. alterniflora* in two study sites (see Fig. 8), *S. alterniflora* was mainly distributed in these regions, such as the frontiers and sparse areas of mangrove forests, and the forest gaps and tidal creeks among the mangroves. Parts of newborn *S. alterniflora* emerged in the form of small and fragmented patches on the mudflats, then the patches gradually joined on the mudflats. Subsequently, *S. alterniflora* further grew seawards or invaded landward vegetation, such as mangrove forests. Besides, the re-emergence of *S. alterniflora* was another pattern of expansion. It has been reported that *S. alterniflora* was cleared from the west coast of the Dandou Sea and then planted in mangrove forests in 2019 [74], while it began to grow and spread in this region again by 2020. The phenomenon of the regrowth of *S. alterniflora* was found in a previous study [75].

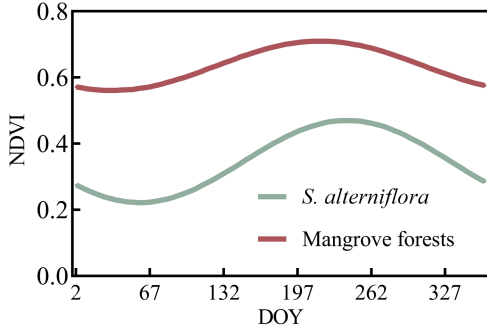


Fig. 9. Phenological curves of *S. alterniflora* and mangrove forests derived from Sentinel-2 NDVI time series.

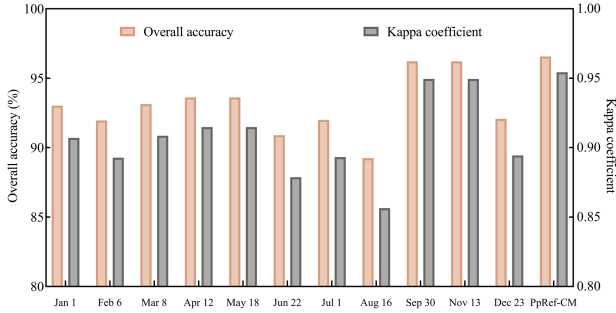


Fig. 10. Comparisons on *S. alterniflora* extraction accuracies using the single-image-based method (single image of each month) and the phenological-based method (image time series of one year).

V. DISCUSSION

A. Impacts of Phenological Information on *S. alterniflora* Extraction

Using the phenological curves derived from Sentinel-2 NDVI time series (see Fig. 9), this study analyzed the phenological differences between *S. alterniflora* and mangrove forests. Overall, the NDVI value of *S. alterniflora* was lower than that of mangrove forests. Besides, the fluctuation of *S. alterniflora* phenological curve was more obvious when compared with evergreen mangrove forests. These phenological differences can help separate *S. alterniflora* from mangrove forests, which is in accordance with an existing study [29].

Considering that the phenology changes of *S. alterniflora* generally cause the spectral variation of vegetation canopy [46], this study further analyzed the influence of phenological information on *S. alterniflora* extraction. Taking the Dandou Sea as an example, the extraction accuracies of *S. alterniflora* in this region were obtained by using the phenological-based method and single-image-based method, respectively (see Fig. 10). Specifically, the phenological-based method chose a series of multiphase images of one year as input, while the single-image-based method selected an image of a specific date for each month as input. As shown in Fig. 10, the proposed PpRef-CM method achieved the highest accuracy of 96.57% and the Kappa coefficient of 0.9543, which outperforms the extraction accuracies of *S. alterniflora* for single-phase image in each month (e.g., 13 November). The former method outperformed the latter in the extraction

of *S. alterniflora*, indicating the effectiveness of phenological information. This result is also consistent with previous studies [25], [31], [76]. In addition, the extraction accuracies of the single-image-based method fluctuated during summer, i.e., June and August. The rainy and cloudy weathers in summer significantly reduce the number of available images [76]. The proposed phenological-based method can take advantage of the phenological information in summer by minimizing the negative effects of cloud cover.

Compared with the single-image-based method, the phenological-based method could capture phenological variability and temporal dynamics of heterogeneous objects [77], [78]. The single-image-based method is susceptible to imaging quality and imaging time when used for extracting *S. alterniflora* due to the complex situations in coastal wetlands, such as the composition of vegetation communities, and periodic tides [10], [25].

B. Applicability of Red-Edge Information for Extracting *S. alterniflora*

The *S. alterniflora* extraction accuracies of the Dandou Sea (Region 1) and the Yingluo Bay (Region 2) in 2020 obtained by using the PpRef-CM and the Ppf-CM [31] were listed in Table VII. Compared with the Ppf-CM, the PpRef-CM showed better performance on *S. alterniflora* extraction with accuracy increased by 2.72% and 2.61% in two regions, respectively.

S. alterniflora tends to occupy empty ecological niches, such as mudflats [79], which makes it hard to distinguish new-born *S. alterniflora* from mudflats, especially for very low vegetation [80]. Besides, although it is easier to extract *S. alterniflora* from mudflats in the GS [81], the misclassification of *S. alterniflora* and mudflats remains. To visually analyze the effectiveness of the red-edge bands (see Fig. 11), Fig. 11(a)–(c) shows the distributions of *S. alterniflora* in the Dandou Sea obtained by using the PpRef-CM and existing Ppf-CM [31]. The PpRef-CM showed better performance than the Ppf-CM in identifying small patches of *S. alterniflora*, which would confuse *S. alterniflora* with mudflats. Moreover, the PpRef-CM proved to be effective in monitoring *S. alterniflora* invasion in the early stages, owing to the advantages of red-edge information in separating *S. alterniflora* from mudflats.

Comparing the extraction results of *S. alterniflora* in the Dandou Sea (Region 1) and the Yingluo Bay (Region 2), the generalization ability of the proposed PpRef-CM was further verified. When compared with the Ppf-CM, the PpRef-CM could recognize more seaward *S. alterniflora*. Moreover, the PpRef-CM could significantly reduce the misclassification between *S. alterniflora* and mudflats [see Fig. 11(e) and (f)], and it improved the OA by 2.61%.

The validity of these classification features (see Table II) was analyzed by using the Shapley additive exPlanations (SHAP). The SHAP is a unified framework based on coalitional game theory, which has been widely used to evaluate the outputs of a prediction model by assigning a score of importance [82], [83]. Fig. 12 shows the SHAP values of different classification features used for *S. alterniflora* extraction with the XGBoost.

TABLE VII
EXTRACTION ACCURACIES OF *S. ALTERNIFLORA* IN TWO STUDY SITES USING THE PROPOSED PpREF-CM AND THE Ppf-CM

Study site	Region 1				Region 2			
Method	Ppf-CM		PpRef-CM		Ppf-CM		PpRef-CM	
Class	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)
<i>S. alterniflora</i>	89.05	91.67	93.81	95.63	91.12	92.42	97.66	96.31
Mangrove forests	94.98	99.05	97.26	99.07	95.85	96.30	98.16	99.53
Mudflats	90.69	84.47	94.61	90.19	91.13	89.81	94.58	94.58
Water	98.12	98.12	98.12	99.05	99.03	98.55	97.09	97.09
OA (%)	93.26		95.98		94.29		96.90	
Kappa coefficient	0.9102		0.9464		0.9238		0.9587	

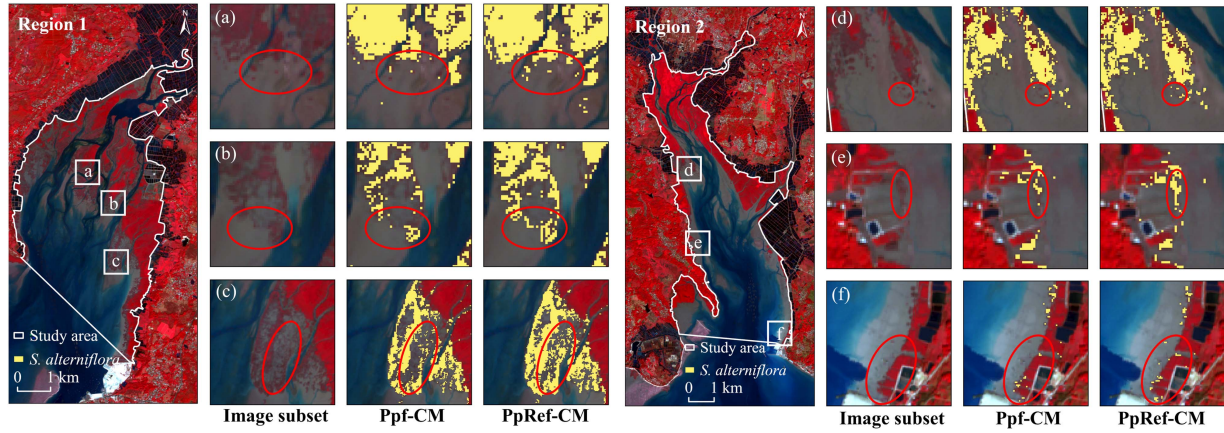


Fig. 11. Comparisons of *S. alterniflora* extraction results in the Dandou Sea (Region 1) and the Yingluo Bay (Region 2) by using the PpRef-CM and existing Ppf-CM.

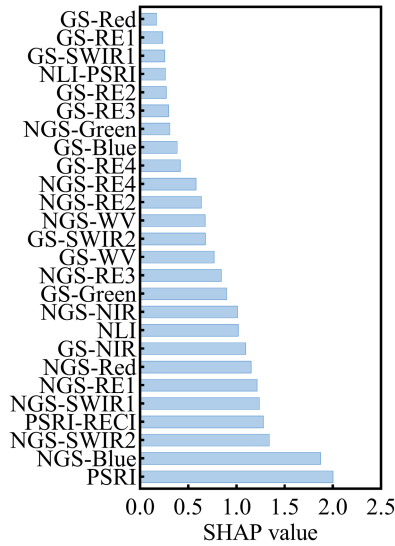


Fig. 12. SHAP value of different classification features in extracting *S. alterniflora* with the XGBoost. A higher SHAP value indicates more contribution to the classification.

Compared with other classification features, the SHAP values of three red-edge band indices (PSRI, PSRI-RECI, and NLI) and two red-edge bands (NGS-RE1 and NGS-RE3) ranked high, demonstrating that the red-edge information of Sentinel-2 can further increase the separability of *S. alterniflora*.

VI. CONCLUSION

To accurately and effectively detect the invasion of *S. alterniflora*, this study developed the PpRef-CM method by combining the phenological information and red-edge bands with Sentinel-2 time series based on the existing Ppf-CM method. The proposed PpRef-CM method and machine-learning algorithms were employed to extract the distributions of *S. alterniflora* in two typical mangrove forests along coastal China. The main conclusions were as follows.

- 1) Integrating the phenological information can clearly increase the extraction accuracies of *S. alterniflora*. As one phenological-based method, the PpRef-CM provided the highest *S. alterniflora* extraction accuracy of 96.57%, when compared with those of single-phase image for all months.
- 2) The red-edge bands and red-edge spectral indices derived from the PpRef-CM-based Sentinel-2 image highlight their advantages in extracting *S. alterniflora*. Compared with the Ppf-CM, by incorporating the red-edge information, the PpRef-CM showed better performance with the *S. alterniflora* extraction accuracies increased by up to 2.72% and 2.61% in two study sites than using only those common spectral bands.
- 3) The XGBoost and RF ensemble learning algorithms outperformed CART and SVM, and the XGBoost gave the best performance on extracting *S. alterniflora* based on all the classification features with the overall accuracies

exceeding 94% in each year and the highest accuracy of 96.57%.

The finding of this study provides the accurate and effective approach for *S. alterniflora* invasion monitoring, and it also contributes to the scientific management of *S. alterniflora* and the ecological security of the coastal zone.

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